



1.01 Complete Vacuum System

1. Purpose

The purpose of the Standard Operating Procedure is to identify the hazards and define the steps for the Vacuum System Operation

2. Scope

This procedure is intended for use by all Silverpoint employees .

3. Hazard Identification

- Equipment Damage and Failure .
- Personal Injury .
- Physical Stress .
- Mental Stress
- Restricted Visibility .
- Pinch Points .
- Diesel Fumes
- Slips, Trips and Falls .
- Burns .
- Hot Pipe Surfaces
- Extreme Noise

4. EHS Requirements

- Fire Retardant Coveralls .
- Steel Toed Boots .
- Safety Glasses .
- Gloves .
- Hard Hat .
- Hearing Protection .
- Proper Labeling of Valves



5. References

- Silverpoint Corporate Codes of Practice
- Alberta OH&S
- Silverpoint Task List
- Silverpoint THA 1.01

6. Procedure

6.1. Vacuum System – the entire vacuum system relies on hydraulics to operate .

6.1.1. On a daily basis you must check your hydraulic tank for the proper oil level and always look at the color of the oil to check for contamination ,

6.1.1.1 . If it is a White Color then there is water in the system and the truck will need to be serviced before it can continue working .

6.1.1.2 . Dark or Black oil indicates a serious problem with hydraulics , contact the maintenance department

6.1.2. When operating the hydraulics in cold weather the hydraulic system must be warmed up before use, if the hydraulic system is cold and you run the truck at the rated RPM it can cause hydraulic hose failure, hydraulic pump damage or damage to the splines where the PTO and the pump join. To warm the system engage the PTO and leave the truck at low idle for 10-15 minutes (Longer in extreme cold) to circulate the oil.

6.1.3. Cold hydraulic oil can cause the vacuum tank to creep up when the pump is first engaged . If this should happen make sure that there is nothing above the tank and make sure the spread plate is off. Normally it only takes a few minutes to warm up enough for it to stop rising, if it doesn't stop, you may need to disengage the hydraulics and try lowering the tank. If you are having problems with getting the tank down call your supervisor .

6.1.4. Hydraulic oil temperature : check the hydraulic oil temperature on a regular basis while operating the truck, there will be a temperature gauge on the hydraulic oil tank. Normal operating temperature is approximately 50 degrees C or 120 degrees F, if you see temperatures above 60 degrees C or 140 degrees F contact your supervisor immediately .

6.1.5. Hydraulic cooling fans: use the manual override switch in the cab to check the operation of the hydraulic cooling fan, make sure the cooling fan is working properly to keep the oil temperature in the proper range .

6.1.6. During hot weather manually turn the fan on so the system can run as cool as possible .

6.2 **Loading** - After Vacuum Truck is in position , then get out and place the chock blocks and check that the blower switch is off and all rear valves are closed . Connect the hose to the load valve on back of truck . When loading Vacuum Truck the four-way valve handle should be pointed to Vacuum ; this should be clearly labeled if not labeled then make arrangements to label it. If you're not sure don't hesitate to call the shop .



6.2.1 Refer to Connecting and disconnecting hoses SOP 1.02, 6.1.1.

1..1.1. Determine where you will be loading from: Tank, Sump?

1..1.2. Determine length and how many hoses you will need to load.

1..1.3. Connect first hose to truck (make sure Vac Unit is in Vacuum Mode)

1..1.4. Connect second hose and more if required.

1..1.5. Your last connection would be the coupler on the tank or the placement of the hose into a tank or sump.

6.2.2 When working with the PTO, make sure that you have 90 psi of air pressure or greater before engaging PTO.

6.2.3 FAILURE TO HAVE THE PROPER AIR PRESSURE BEFORE ENGAGING PTO WILL RESULT IN THE PTO TRYING TO ENGAGE ITSELF ONCE THE PRESSURE HAS REACHED 90PSI, THIS WILL FORCE THE PTO GEAR IN TO THE TRANSMISSION GEARS AND THEY WILL NOT ENGAGE, THE GEARS WILL GRIND ON EACH OTHER UNTIL THEY ARE DESTROYED !

6.2.4 When engaging the PTO, push the clutch all the way to the floor and put the transmission into gear, engage PTO, take the transmission out of gear and slowly release the clutch. The PTO should not be making any grinding noises, if it is, push the clutch in immediately and start again.

6.2.5 When working with an automatic transmission, PUT TRANSMISSION IN NEUTRAL, SET THE PARK BRAKE, AT AN IDLE, turn the PTO switch to ON. If you do not put the transmission in neutral, when you idle the engine up it will cause the truck to move.

6.2.6 Set specified RPM for that particular unit blower operation (Labeled in cab of truck). Exit the cab, connect the hoses. (Follow Connecting and Disconnecting hoses procedure 1.11). Ensure four way valve is on "vacuum", turn on the HIBON, then open the load valve.

6.2.7 When disengaging the PTO, Idle truck down, push the clutch in all the way to the floor and put the transmission into gear, then switch off the PTO. Most damage to the PTO is done when disengaging, because there is still pressure on the gears if you do not have the clutch in. When disengaging the PTO with an automatic transmission, IDLE DOWN before shutting it off.

6.3 Hibon Gearbox Oil Level - There are two sight glasses on your vacuum pump. One is located at the gearbox drive shaft end and the other at the free end. The oil level is correct when it reaches half way or three quarters up the sight glass. You should check the level when the pump is at rest at the start of the day and periodical throughout the day. If the oil is milky looking before operation then you will have to change oil and record on the PMI.

6.4 Hibon Oil Changes - The initial oil change should be done at 200 hours of operation and further oil changes should be conducted at 300 hours. Oil should be replaced with synthetic 50W oil. **Remember that there are two sides to change.**

6.5 Flushing Hibon System - If you are not going to use the vacuum truck for a while then make sure to diesel flush the system. A small amount of diesel fuel should be used to flush the system. Do not use excessive amounts, hot temperatures and diesel fuel don't mix could cause a flash fire. Flushing is required more often in the cold weather to



help prevent the pump from freezing This also clears out debris that may have gathered in the pump itself . I.e.: cement , ice and mud .

6.6 Refer to Vacuum /Pressure diagrams and vacuum system diagnostic charts . Learn the parts of the system : vacuum pump , cyclone , filters , float ball , gauge , 4 way valve , silencers , pressure /vacuum , and check valves .

6.7 **Pressure** – Do not use pressure to unplug vacuum hoses , the volume of the tank is so great that it may blow the hose apart , or dislodge whatever is in the hose , and possibly spray all over the rig. This could spray someone as well and cause an injury . To relieve pressure in the tank move the Vacuum pressure lever slowly to the half way position . Never switch from pressure to vacuum without slowly relieving the pressure first. Not relieving first will cause the pump to run backwards and damage the pump or pop the shut off ball inside the tank up into the piping causing pressure to be trapped inside the tank . If this happens **Contact your Supervisor**

6.8 **Vacuum** – Used to load , burp and to clear out 6 inch valve . Ear plugs must be worn when the pump is in operation because of the constant high pitch noise of the pump . Refer to diagram in the operations manual .

6.9 **Controls** – 4 way valve , hoist , vibrator , agitator , diesel flush , and spread valve .

6.10 **Cyclone , Filter and Valves** - Cleaning out valves after cement and every load is a must . Draining your cyclone and silencers after every load also helps in allowing the whole system to run better at full vacuum and pressure . Cleaning your final filter on a daily basis is a must . When you seem to lose vacuum , the first place that you should check is the final filter , if it is dirty or plugged , this will cause you to lose vacuum or pressure . Understanding how to do a vacuum build up and leak down test , is a good trouble shooting method to finding , leaking valves or other potential problems .

Out of Service – If the final filter and cyclone are not drained and cleaned every day and or after every load The Vacuum unit will be considered Out of Service until such time as this procedure has been performed .

6.11 **Tank** – Inside the tank there are two components that are very important to the operation of the whole system . The tank gauge and shut off ball . The tank gauge tells you how much fluid you have in the tank . The gauge will tend to freeze to the bottom of the tank in the winter . If you keep your tank clean , you can avoid this . In extreme cold weather the gauge has also known to become stiff , just keep this in mind when working in these conditions .

6.12 **Shut Off Ball** -The shut off ball is inside the tank and is there to stop fluid from going through the vacuum system when the tank gets full . When the tank is full and the ball floats up it seats into a rubber seal and stops the vacuum system from building any more vacuum in the tank .

DO NOT rely on the shut off ball to gauge how full your tank is . If you do this , it will cause problems out in the field when you try and spread a load . Some things to check on the shut off ball are , shape of the ball , if the hanger for the ball is in good shape and attached , the rubber ring that the ball seats in , and whether the ball moves freely up and down the hanger

6.13 **Tank Safety Bar** – Every truck is equipped with a safety bar . This is put into place so that when you are in the tank , the rear door cannot close . If the bar has been lost , get a replacement ; do not go into the tank without one .

6.14 **Tank Hoist Lockout Device** - This safety device is used for the hoist , to prevent the tank from lowering in case a hydraulic



line should fail while a worker is underneath the tank during service or repairs .

6.15 Hydraulic Safety ; DO NOT RUN THE HYDRAULICS ABOVE THE RATED RPM, SEVERE SYSTEM DAMAGE WILL RESULT. NOT MAYBE . NOT AFTER AWHILE . BUT IMMEDIATE SYSTEM DAMAGE WILL RESULT.

6.16 Hoses , Cam locks and Caps – Inspect hoses to be in good condition no holes tears deep abrasions . Each end; the male and female end require two bands each to secure the hose to each nipple . The female end; the cams must move easily and cannot be worn bent or other .

Out of Service - If a hose does not meet the proceeding condition it must be placed Out of Service .

6.17 Remember that these units are a wet vacuum system . Do not load dry material ; wet it down before loading it (such as dry cement , dirt, saw dust). Loading dry material will allow particles of the dry material to flow though the entire vacuum system and into the Hibon pump . These pumps are designed to run very tight tolerances , any material that flows thru the will damage the pump .

Remember : If you are in doubt about what to do, call your supervisor .